

Statics of Bending: Shear and Bending Moment Diagrams

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Introduction

Beams are long and slender structural elements, differing from truss elements in that they are called on to support transverse as well as axial loads. Their attachment points can also be more complicated than those of truss elements: they may be bolted or welded together, so the attachments can transmit bending moments or transverse forces into the beam. Beams are among the most common of all structural elements, being the supporting frames of airplanes, buildings, cars, people, and much else.

The nomenclature of beams is rather standard: as shown in Fig. 1, L is the length, or span; b is the width, and h is the height (also called the *depth*). The cross-sectional shape need not be rectangular, and often consists of a vertical *web* separating horizontal *flanges* at the top and bottom of the beam¹.

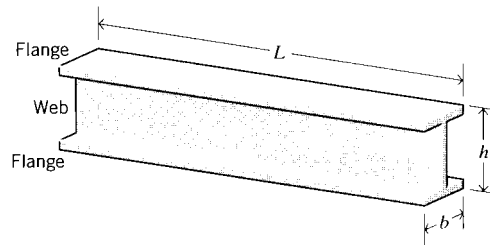


Figure 1: Beam nomenclature.

As will be seen in Modules 13 and 14, the stresses and **deflections** induced in a beam under bending loads vary along the beam's length and height. The first step in calculating these quantities and their spatial variation consists of constructing *shear* and *bending moment* diagrams, $V(x)$ and $M(x)$, which are the internal shearing forces and bending moments induced in the beam, plotted along the beam's length. The following sections will describe how these diagrams are made.

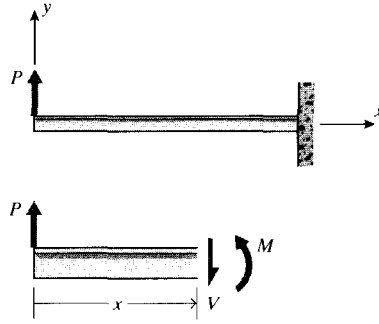


Figure 2: A cantilevered beam.

Free-body diagrams

As a simple starting example, consider a beam clamped (“cantilevered”) at one end and subjected to a load P at the free end as shown in Fig. 2. A free body diagram of a section cut transversely at position x shows that a shear force V and a moment M must exist on the cut section to maintain equilibrium. We will show in Module 13 that these are the resultants of shear and normal stresses that are set up on internal planes by the bending loads. As usual, we will consider section areas whose normals point in the $+x$ direction to be positive; then shear forces pointing in the $+y$ direction on $+x$ faces will be considered positive. Moments whose vector direction as given by the right-hand rule are in the $+z$ direction (vector out of the plane of the paper, or tending to cause counterclockwise rotation in the plane of the paper) will be positive when acting on $+x$ faces. Another way to recognize positive bending moments is that they cause the bending shape to be concave upward. For this example beam, the statics equations give:

$$\sum F_y = 0 = V + P \Rightarrow V = \text{constant} = -P \quad (1)$$

$$\sum M_0 = 0 = -M + Px \Rightarrow M = M(x) = Px \quad (2)$$

Note that the moment increases with distance from the loaded end, so the magnitude of the maximum value of M compared with V increases as the beam becomes longer. This is true of most beams, so shear effects are usually more important in beams with small length-to-height ratios.

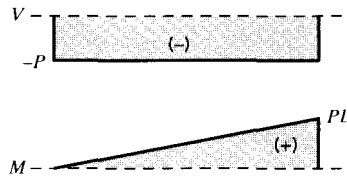


Figure 3: Shear and bending moment diagrams.

¹There is a standardized protocol for denoting structural steel beams; for instance W 8 × 40 indicates a wide-flange beam with a nominal depth of 8" and weighing 40 lb/ft of length

As stated earlier, the stresses and deflections will be shown to be functions of V and M , so it is important to be able to compute how these quantities vary along the beam's length. Plots of $V(x)$ and $M(x)$ are known as *shear* and *bending moment diagrams*, and it is necessary to obtain them before the stresses can be determined. For the end-loaded cantilever, the diagrams shown in Fig. 3 are obvious from Eqns. 1 and 2.

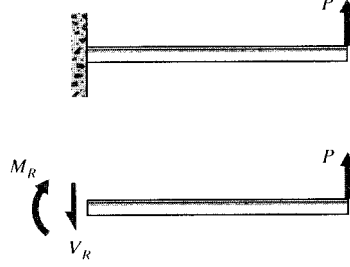


Figure 4: Wall reactions for the cantilevered beam.

It was easiest to analyze the cantilevered beam by beginning at the free end, but the choice of origin is arbitrary. It is not always possible to guess the easiest way to proceed, so consider what would have happened if the origin were placed at the wall as in Fig. 4. Now when a free body diagram is constructed, forces must be placed at the origin to replace the reactions that were imposed by the wall to keep the beam in equilibrium with the applied load. These reactions can be determined from free-body diagrams of the beam as a whole (if the beam is statically determinate), and must be found before the problem can proceed. For the beam of Fig. 4:

$$\sum F_y = 0 = -V_R + P \Rightarrow V_R = P$$

$$\sum M_o = 0 = M_R - PL \Rightarrow M_R = PL$$

The shear and bending moment at x are then

$$V(x) = V_R = P = \text{constant}$$

$$M(x) = M_R - V_R x = PL - Px$$

This choice of origin produces some extra algebra, but the $V(x)$ and $M(x)$ diagrams shown in Fig. 5 are the same as before (except for changes of sign): V is constant and equal to P , and M varies linearly from zero at the free end to PL at the wall.

Distributed loads

Transverse loads may be applied to beams in a distributed rather than at-a-point manner as depicted in Fig. 6, which might be visualized as sand piled on the beam. It is convenient to describe these distributed loads in terms of *force per unit length*, so that $q(x) dx$ would be the load applied to a small section of length dx by a distributed load $q(x)$. The shear force $V(x)$ set up in reaction to such a load is

$$V(x) = - \int_{x_0}^x q(\xi) d\xi \quad (3)$$

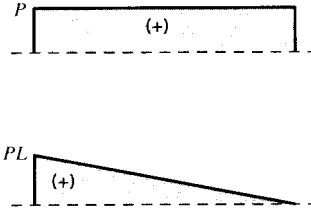


Figure 5: Alternative shear and bending moment diagrams for the cantilevered beam.

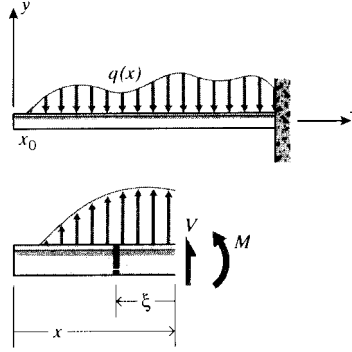


Figure 6: A distributed load and a free-body section.

where x_0 is the value of x at which $q(x)$ begins, and ξ is a dummy length variable that looks backward from x . Hence $V(x)$ is the area under the $q(x)$ diagram up to position x . The moment balance is obtained considering the increment of load $q(\xi) d\xi$ applied to a small width $d\xi$ of beam, a distance ξ from point x . The incremental moment of this load around point x is $q(\xi) \xi d\xi$, so the moment $M(x)$ is

$$M = \int_{x_0}^x q(\xi) \xi d\xi \quad (4)$$

This can be related to the centroid of the area under the $q(x)$ curve up to x , whose distance from x is

$$\bar{\xi} = \frac{\int q(\xi) \xi d\xi}{\int q(\xi) d\xi}$$

Hence Eqn. 4 can be written

$$M = Q\bar{\xi} \quad (5)$$

where $Q = \int q(\xi) d\xi$ is the area. Therefore, the distributed load $q(x)$ is statically equivalent to a concentrated load of magnitude Q placed at the centroid of the area under the $q(x)$ diagram.

Example 1

Consider a simply-supported beam carrying a triangular and a concentrated load as shown in Fig. 7. For

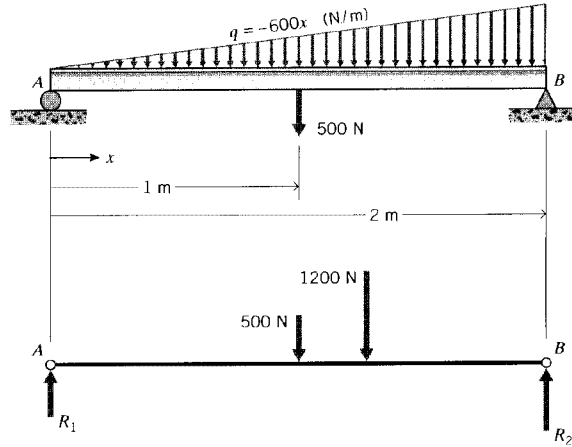


Figure 7: Distributed and concentrated loads.

the purpose of determining the support reaction forces R_1 and R_2 , the distributed triangular load can be replaced by its static equivalent. The magnitude of this equivalent force is

$$Q = \int_0^2 (-600x) dx = -1200$$

The equivalent force acts through the centroid of the triangular area, which is $2/3$ of the distance from its narrow end (see Prob. 1). The reaction R_2 can now be found by taking moments around the left end:

$$\sum M_A = 0 = -500(1) - (1200)(2/3) + R_2(2) \rightarrow R_2 = 650$$

The other reaction can then be found from vertical equilibrium:

$$\sum F_y = 0 = R_1 - 500 - 1200 + 650 = 1050$$

Successive integration method

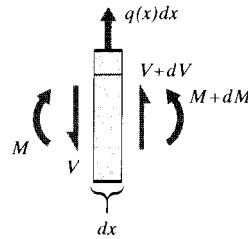


Figure 8: Relations between distributed loads and internal shear forces and bending moments.

We have already noted in Eqn. 3 that the shear curve is the negative integral of the loading curve. Another way of developing this is to consider a free body balance on a small increment

of length dx over which the shear and moment changes from V and M to $V + dV$ and $M + dM$ (see Fig. 8). The distributed load $q(x)$ can be taken as constant over the small interval, so the force balance is:

$$\sum F_y = 0 = V + dV + q dx - V = 0$$

$$\boxed{\frac{dV}{dx} = -q} \quad (6)$$

or

$$\boxed{V(x) = - \int q(x) dx} \quad (7)$$

which is equivalent to Eqn. 3. A moment balance around the center of the increment gives

$$\sum M_o = (M + dM) + (V + dV) \frac{dx}{2} + V \frac{dx}{2} - M$$

As the increment dx is reduced to the limit, the term containing the higher-order differential $dV dx$ vanishes in comparison with the others, leaving

$$\boxed{\frac{dM}{dx} = -V} \quad (8)$$

or

$$\boxed{M(x) = - \int V(x) dx} \quad (9)$$

Hence the value of the shear curve at any axial location along the beam is equal to the negative of the slope of the moment curve at that point, and the value of the moment curve at any point is equal to the negative of the area under the shear curve up to that point.

The shear and moment curves can be obtained by successive integration of the $q(x)$ distribution, as illustrated in the following example.

Example 2

Consider a cantilevered beam subjected to a negative distributed load $q(x) = -q_0 = \text{constant}$ as shown in Fig. 9; then

$$V(x) = - \int q(x) dx = q_0 x + c_1$$

where c_1 is a constant of integration. A free body diagram of a small sliver of length near $x = 0$ shows that $V(0) = 0$, so the c_1 must be zero as well. The moment function is obtained by integrating again:

$$M(x) = - \int V(x) dx = -\frac{1}{2} q_0 x^2 + c_2$$

where c_2 is another constant of integration that is also zero, since $M(0) = 0$.

Admittedly, this problem was easy because we picked one with null boundary conditions, and with only one loading segment. When concentrated or distributed loads are found at different

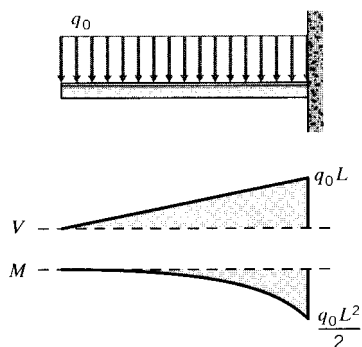


Figure 9: Shear and moment distributions in a cantilevered beam.

positions along the beam, it is necessary to integrate over each section between loads separately. Each integration will produce an unknown constant, and these must be determined by invoking the continuity of slopes and deflections from section to section. This is a laborious process, but one that can be made much easier using *singularity functions* that will be introduced shortly.

It is often possible to sketch V and M diagrams without actually drawing free body diagrams or writing equilibrium equations. This is made easier because the curves are integrals or derivatives of one another, so graphical sketching can take advantage of relations among slopes and areas.

These rules can be used to work gradually from the $q(x)$ curve to $V(x)$ and then to $M(x)$. Wherever a concentrated load appears on the beam, the $V(x)$ curve must jump by that value, but in the opposite direction; similarly, the $M(x)$ curve must jump discontinuously wherever a couple is applied to the beam.

Example 3

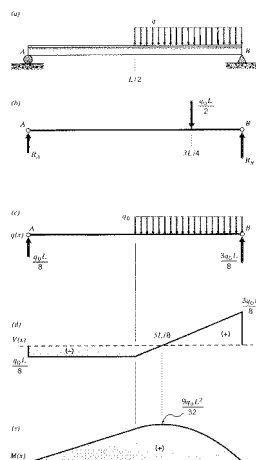


Figure 10: A simply supported beam.

To illustrate this process, consider a simply-supported beam of length L as shown in Fig. 10, loaded

over half its length by a negative distributed load $q = -q_0$. The solution for $V(x)$ and $M(x)$ takes the following steps:

1. The reactions at the supports are found from static equilibrium. Replacing the distributed load by a concentrated load $Q = -q_0(L/2)$ at the midpoint of the q distribution (Fig. 10(b)) and taking moments around A :

$$R_B L = \left(\frac{q_0 L}{2} \right) \left(\frac{3L}{4} \right) \Rightarrow R_B = \frac{3q_0 L}{8}$$

The reaction at the right end is then found from a vertical force balance:

$$R_A = \frac{q_0 L}{2} - R_B = \frac{q_0 L}{8}$$

Note that only two equilibrium equations were available, since a horizontal force balance would provide no relevant information. Hence the beam will be statically indeterminate if more than two supports are present.

The $q(x)$ diagram is then just the beam with the end reactions shown in Fig. 10(c).

2. Beginning the shear diagram at the left, V immediately jumps down to a value of $-q_0 L/8$ in opposition to the discontinuously applied reaction force at A ; it remains at this value until $x = L/2$ as shown in Fig. 10(d).
3. At $x = L/2$, the $V(x)$ curve starts to rise with a constant slope of $+q_0$ as the area under the $q(x)$ distribution begins to accumulate. When $x = L$, the shear curve will have risen by an amount $q_0 L/2$, the total area under the $q(x)$ curve; its value is then $(-q_0 L/8) + (q_0 L/2) = (3q_0 L/8)$. The shear curve then drops to zero in opposition to the reaction force $R_B = (3q_0 L/8)$. (The V and M diagrams should always close, and this provides a check on the work.)
4. The moment diagram starts from zero as shown in Fig. 10(e), since there is no discontinuously applied moment at the left end. It moves upward at a constant slope of $+q_0 L/8$, the value of the shear diagram in the first half of the beam. When $x = L/2$, it will have risen to a value of $q_0 L^2/16$.
5. After $x = L/2$, the slope of the moment diagram starts to fall as the value of the shear diagram rises. The moment diagram is now parabolic, always being one order higher than the shear diagram. The shear diagram crosses the $V = 0$ axis at $x = 5L/8$, and at this point the slope of the moment diagram will have dropped to zero. The maximum value of M is $9q_0 L^2/32$, the total area under the V curve up to this point.
6. After $x = 5L/8$, the moment diagram falls parabolically, reaching zero at $x = L$.

Singularity functions

This special family of functions provides an automatic way of handling the irregularities of loading that usually occur in beam problems. They are much like conventional polynomial factors, but with the property of being zero until “activated” at desired points along the beam. The formal definition is

$$f_n(x) = \langle x - a \rangle^n = \begin{cases} 0, & x < a \\ (x - a)^n, & x > a \end{cases} \quad (10)$$

where $n = -2, -1, 0, 1, 2, \dots$. The function $\langle x - a \rangle^0$ is a unit step function, $\langle x - a \rangle_{-1}$ is a concentrated load, and $\langle x - a \rangle_{-2}$ is a concentrated couple. The first five of these functions are sketched in Fig. 11.

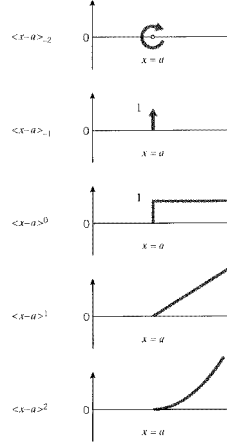


Figure 11: Singularity functions.

The singularity functions are integrated much like conventional polynomials:

$$\int_{-\infty}^x \langle x-a \rangle^n dx = \frac{\langle x-a \rangle^{n+1}}{n+1} \quad n \geq 0 \quad (11)$$

However, there are special integration rules for the $n = -1$ and $n = -2$ members, and this special handling is emphasized by using subscripts for the n index:

$$\int_{-\infty}^x \langle x-a \rangle_{-2} dx = \langle x-a \rangle_{-1} \quad (12)$$

$$\int_{-\infty}^x \langle x-a \rangle_{-1} dx = \langle x-a \rangle^0 \quad (13)$$

Example 4

Applying singularity functions to the beam of Example 4.3, the loading function would be written

$$q(x) = +\frac{q_0 L}{8} \langle x-0 \rangle_{-1} - q_0 \langle x - \frac{L}{2} \rangle^0$$

The reaction force at the right end could also be included, but it becomes activated only as the problem is over. Integrating once:

$$V(x) = - \int q(x) dx = -\frac{q_0 L}{8} \langle x \rangle^0 + q_0 \langle x - \frac{L}{2} \rangle^1$$

The constant of integration is included automatically here, since the **influence** of the reaction at A has been included explicitly. Integrating again:

$$M(x) = - \int V(x) dx = \frac{q_0 L}{8} \langle x \rangle^1 - \frac{q_0}{2} \langle x - \frac{L}{2} \rangle^2$$

Examination of this result will show that it is the same as that developed previously.

MapleTM symbolic manipulation software provides an efficient means of plotting these functions. The following shows how the moment equation of this example might be plotted, using the Heaviside function to provide the singularity.

```

# Define function sfm in terms of a and n
>sfm:=proc(a,n) (x-a)^n*Heaviside(x-a) end;

      sfm := proc(a, n) (x - a)^n*Heaviside(x - a) end proc

# Input moment equation using singularity functions
>M(x):=(q*L/8)*sfm(0,1)-(q/2)*sfm(L/2,2);

      M(x) := 1/8 q L x Heaviside(x)
              2
              - 1/2 q (x - 1/2 L) Heaviside(x - 1/2 L)

# Provide numerical values for q and L:
>q:=1: L:=10:

# Plot function
>plot(M(x),x=0..10);

```

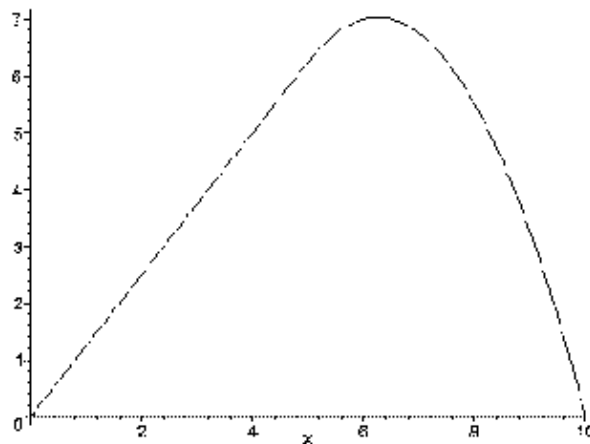
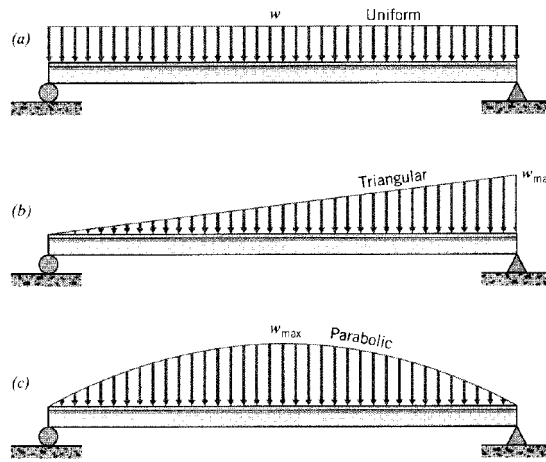


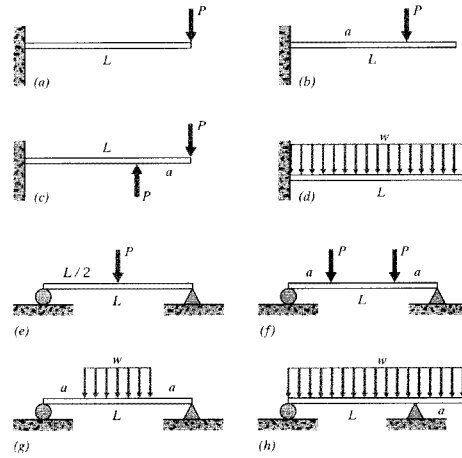
Figure 12: Maple singularity plot

Problems

1. (a)–(c) Locate the magnitude and position of the force equivalent to the loading distributions shown here.
2. (a)–(c) Determine the reaction forces at the supports of the cases in Prob. 1.
3. (a)–(h) Sketch the shear and bending moment diagrams for the load cases shown here.
4. (a)–(h) Write singularity-function expressions for the shear and bending moment distributions for the cases in Prob. 3.
5. (a)–(h) Use Maple (or other) software to plot the shear and bending moment distributions for the cases in Prob. 3, using the values (as needed) $L = 25$ in, $a = 5$ in, $w = 10$ lb/in, $P = 150$ lb.

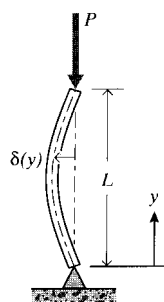


Prob. 1

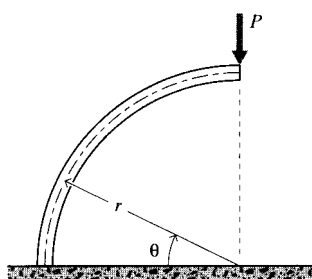


Prob. 3

6. The transverse deflection of a beam under an axial load P is taken to be $\delta(y) = \delta_0 \sin(y\pi/L)$, as shown here. Determine the bending moment $M(y)$ along the beam.
7. Determine the bending moment $M(\theta)$ along the circular curved beam shown.



Prob. 6



Prob. 7

STRESSES IN BEAMS

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Introduction

Understanding of the stresses induced in beams by bending loads took many years to develop. Galileo worked on this problem, but the theory as we use it today is usually credited principally to the great mathematician Leonard Euler (1707–1783). As will be developed below, beams develop normal stresses in the lengthwise direction that vary from a maximum in tension at one surface, to zero at the beam’s midplane, to a maximum in compression at the opposite surface. Shear stresses are also induced, although these are often negligible in comparison with the normal stresses when the length-to-height ratio of the beam is large. The procedures for calculating these stresses for various loading conditions and beam cross-section shapes are perhaps the most important methods to be found in introductory Mechanics of Materials, and will be developed in the sections to follow. This theory requires that the user be able to construct shear and bending moment diagrams for the beam, as developed for instance in Module 12.

Normal Stresses

A beam subjected to a positive bending moment will tend to develop a concave-upward curvature. Intuitively, this means the material near the top of the beam is placed in compression along the x direction, with the lower region in tension. At the transition between the compressive and tensile regions, the stress becomes zero; this is the *neutral axis* of the beam. If the material tends to fail in tension, like chalk or glass, it will do so by crack initiation and growth from the lower tensile surface. If the material is strong in tension but weak in compression, it will fail at the top compressive surface; this might be observed in a piece of wood by a compressive buckling of the outer fibers.

We seek an expression relating the magnitudes of these axial normal stresses to the shear and bending moment within the beam, analogously to the shear stresses induced in a circular shaft by torsion. In fact, the development of the needed relations follows exactly the same direct approach as that used for torsion:

1. *Geometrical statement:* We begin by stating that originally transverse planes within the beam remain planar under bending, but rotate through an angle θ about points on the neutral axis as shown in Fig. 1. For small rotations, this angle is given approximately by

the x -derivative of the beam's vertical deflection function $v(x)$ ¹:

$$u = -yv_{,x} \quad (1)$$

where the comma indicates differentiation with respect to the indicated variable ($v_{,x} \equiv dv/dx$). Here y is measured positive upward from the neutral axis, whose location within the beam has not yet been determined.

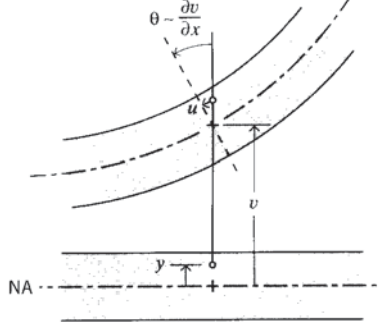


Figure 1: Geometry of beam bending.

2. *Kinematic equation:* The x -direction normal strain ϵ_x is then the gradient of the displacement:

$$\epsilon_x = \frac{du}{dx} = -yv_{,xx} \quad (2)$$

Note that the strains are zero at the neutral axis where $y = 0$, negative (compressive) above the axis, and positive (tensile) below. They increase in magnitude linearly with y , much as the shear strains increased linearly with r in a torsionally loaded circular shaft. The quantity $v_{,xx} \equiv d^2v/dx^2$ is the spatial rate of change of the slope of the beam deflection curve, the “slope of the slope.” This is called the *curvature* of the beam.

3. *Constitutive equation:* The stresses are obtained directly from Hooke’s law as

$$\sigma_x = E\epsilon_x = -yEv_{,xx} \quad (3)$$

This restricts the applicability of this derivation to linear elastic materials. Hence the axial normal stress, like the strain, increases linearly from zero at the neutral axis to a maximum at the outer surfaces of the beam.

4. *Equilibrium relations:* Since there are no axial (x -direction) loads applied externally to the beam, the total axial force generated by the normal σ_x stresses (shown in Fig. 2) must be zero. This can be expressed as

¹The exact expression for curvature is

$$\frac{d\theta}{ds} = \frac{d^2v/dx^2}{[1 + (dv/dx)^2]^{3/2}}$$

This gives $\theta \approx dv/dx$ when the squared derivative in the denominator is small compared to 1.

$$\sum F_x = 0 = \int_A \sigma_x dA = \int_A -yEv_{,xx} dA$$

which requires that

$$\int_A y dA = 0$$

The distance $\bar{y}$ from the neutral axis to the centroid of the cross-sectional area is

$$\bar{y} = \frac{\int_A y dA}{\int_A dA}$$

Hence $\bar{y} = 0$, i.e. the neutral axis is coincident with the centroid of the beam cross-sectional area. This result is obvious on reflection, since the stresses increase at the same linear rate, above the axis in compression and below the axis in tension. Only if the axis is exactly at the centroidal position will these stresses balance to give zero net horizontal force and keep the beam in horizontal equilibrium.

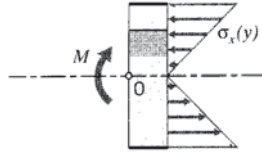


Figure 2: Moment and force equilibrium in the beam.

The normal stresses in compression and tension are balanced to give a zero net horizontal force, but they also produce a net clockwise moment. This moment must equal the value of $M(x)$ at that value of x , as seen by taking a moment balance around point O :

$$\sum M_O = 0 = M + \int_A \sigma_x \cdot y dA$$

$$M = \int_A (yEv_{,xx}) \cdot y dA = Ev_{,xx} \int_A y^2 dA \quad (4)$$

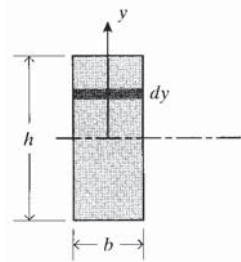


Figure 3: Moment of inertia for a rectangular section.

The quantity $\int y^2 dA$ is the rectangular moment of inertia with respect to the centroidal axis, denoted I . For a rectangular cross section of height h and width b as shown in Fig. 3 this is:

$$I = \int_{-h/2}^{h/2} y^2 b dy = \frac{bh^3}{12} \quad (5)$$

Solving Eqn. 4 for $v_{,xx}$, the beam curvature is

$$v_{,xx} = \frac{M}{EI} \quad (6)$$

5. An explicit formula for the stress can be obtained by using this in Eqn. 3:

$$\sigma_x = -yE \frac{M}{EI} = \frac{-My}{I} \quad (7)$$

The final expression for stress, Eqn. 7, is similar to $\tau_{\theta z} = Tr/J$ for twisted circular shafts: the stress varies linearly from zero at the neutral axis to a maximum at the outer surface, it varies inversely with the moment of inertia of the cross section, and it is independent of the material's properties. Just as a designer will favor annular drive shafts to maximize the polar moment of inertia J , beams are often made with wide flanges at the upper and lower surfaces to increase I .

Example 1

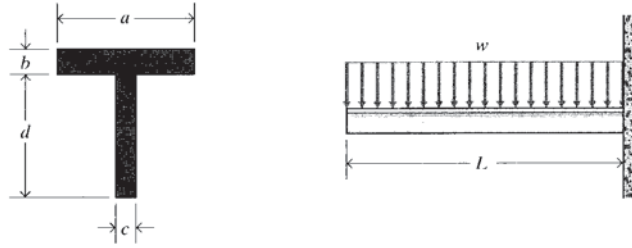


Figure 4: A cantilevered T-beam.

Consider a cantilevered T-beam with dimensions as shown in Fig. 4, carrying a uniform loading of w N/m. The maximum bending moment occurs at the wall, and is easily found to be $M_{max} = (wL)(L/2)$. The stress is then given by Eqn. 7, which requires that we know the location of the neutral axis (since y and I are measured from there).

The distance $\bar{y}$ from the bottom of the beam to the centroidal neutral axis can be found using the “composite area theorem” (see Prob. 1). This theorem states that the distance from an arbitrary axis to the centroid of an area made up of several subareas is the sum of the subareas times the distance to their individual centroids, divided by the sum of the subareas (i.e. the total area):

$$\bar{y} = \frac{\sum_i A_i \bar{y}_i}{\sum_i A_i}$$

For our example, this is

$$\bar{y} = \frac{(d/2)(cd) + (d + b/2)(ab)}{cd + ab}$$

The moments of inertia of the individual parts of the compound area with respect to their own centroids are just $ab^3/12$ and $cd^3/12$. These moments can be referenced to the horizontal axis through the centroid of the compound area using the “parallel axis theorem” (see Prob. 3). This theorem states that the moment of inertia $I_{z'}$ of an area A , relative to any arbitrary axis z' parallel to an axis through the centroid but a distance d from it, is the moment of inertia relative to the centroidal axis I_z plus the product of the area A and the square of the distance d :

$$I_{z'} = I_z + Ad^2$$

For our example, this is

$$I^{(1)} = \frac{ab^3}{12} + (ab) \left(d + \frac{b}{2} - \bar{y} \right)^2$$

$$I^{(2)} = \frac{cd^3}{12} + (cd) \left(\frac{d}{2} - \bar{y} \right)^2$$

The moment of inertia of the entire compound area, relative to its centroid, is then the sum of these two contributions:

$$I = I^{(1)} + I^{(2)}$$

The maximum stress is then given by Eqn. 7 using this value of I and $y = \bar{y}/2$ (the distance from the neutral axis to the outer fibers), along with the maximum bending moment M_{max} . The result of these substitutions is

$$\sigma_x = \frac{(3d^2c + 6abd + 3ab^2)wL^2}{2c^2d^4 + 8abcd^3 + 12ab^2cd^2 + 8ab^3cd + 2a^2b^4}$$

In practice, each step would likely be reduced to a numerical value rather than working toward an algebraic solution.

In pure bending (only bending moments applied, no transverse or longitudinal forces), the only stress is σ_x as given by Eqn. 7. All other stresses are zero ($\sigma_y = \sigma_z = \tau_{xy} = \tau_{xz} = \tau_{yz} = 0$). However, strains other than ϵ_x are present, due to the Poisson effect. This does not generate shear strain ($\gamma_{xy} = \gamma_{xz} = \gamma_{yz} = 0$), but the normal strains are

$$\epsilon_x = \frac{1}{E}[\sigma_x - \nu(\sigma_y + \sigma_z)] = \frac{\sigma_x}{E}$$

$$\epsilon_y = \frac{1}{E}[\sigma_y - \nu(\sigma_x + \sigma_z)] = -\nu \frac{\sigma_x}{E}$$

$$\epsilon_z = \frac{1}{E}[\sigma_z - \nu(\sigma_x + \sigma_y)] = -\nu \frac{\sigma_x}{E}$$

The strains can also be written in terms of curvatures. From Eqn. 2, the curvature along the beam is

$$v_{,xx} = -\frac{\epsilon_x}{y}$$

This is accompanied by a curvature transverse to the beam axis given by

$$v_{,zz} = -\frac{\epsilon_z}{y} = \frac{\nu\epsilon_x}{y} = -\nu v_{,xx}$$

This transverse curvature, shown in Fig. 5, is known as *anticlastic curvature*; it can be seen by bending a “Pink Pearl” type eraser in the fingers.

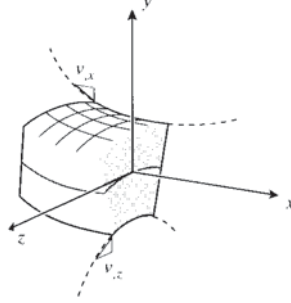


Figure 5: Anticlastic curvature.

As with tension and torsion structures, bending problems can often be done more easily with energy methods. Knowing the stress from Eqn. 7, the strain energy due to bending stress U_b can be found by integrating the strain energy per unit volume $U^* = \sigma^2/2E$ over the specimen volume:

$$\begin{aligned} U_b &= \int_V U^* dV = \int_L \int_A \frac{\sigma_x^2}{2E} dA dL \\ &= \int_L \int_A \frac{1}{2E} \left(\frac{-My}{I} \right)^2 dA dL = \int_L \frac{M^2}{2EI^2} \int_A y^2 dA dL \end{aligned}$$

Since $\int_A y^2 dA = I$, this becomes

$$\boxed{U_b = \int_L \frac{M^2 dL}{2EI}} \quad (8)$$

If the bending moment is constant along the beam (definitely *not* the usual case), this becomes

$$U = \frac{M^2 L}{2EI}$$

This is another analog to the expression for uniaxial tension, $U = P^2 L/2AE$.

Buckling

Long slender columns placed in compression are prone to fail by buckling, in which the column develops a kink somewhere along its length and quickly collapses unless the load is relaxed. This is actually a bending phenomenon, driven by the bending moment that develops if and when the beam undergoes a transverse deflection. Consider a beam loaded in axial compression and pinned at both ends as shown in Fig. 6. Now let the beam be made to deflect transversely by an amount v , perhaps by an adventitious sideward load or even an irregularity in the beam’s cross section. Positions along the beam will experience a moment given by

$$M(x) = Pv(x) \quad (9)$$

The beam's own stiffness will act to restore the deflection and recover a straight shape, but the effect of the bending moment is to deflect the beam *more*. It's a battle over which influence wins out. If the tendency of the bending moment to increase the deflection dominates over the ability of the beam's elastic stiffness to resist bending, the beam will become unstable, continuing to bend at an accelerating rate until it fails.

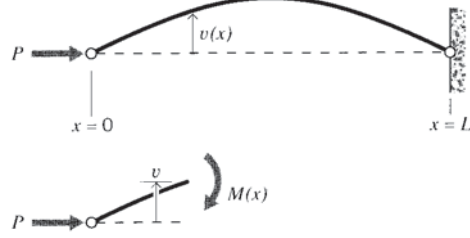


Figure 6: Imminent buckling in a beam.

The bending moment is related to the beam curvature by Eqn. 6, so combining this with Eqn. 9 gives

$$v_{,xx} = \frac{P}{EI} v \quad (10)$$

Of course, this governing equation is satisfied identically if $v = 0$, i.e. the beam is straight. We wish to look beyond this trivial solution, and ask if the beam could adopt a *bent* shape that would also satisfy the governing equation; this would imply that the stiffness is insufficient to restore the unbent shape, so that the beam is beginning to buckle. Equation 10 will be satisfied by functions that are proportional to their own second derivatives. Trigonometric functions have this property, so candidate solutions will be of the form

$$v = c_1 \sin \sqrt{\frac{P}{EI}} x + c_2 \cos \sqrt{\frac{P}{EI}} x$$

It is obvious that c_2 must be zero, since the deflection must go to zero at $x = 0$ and L . Further, the sine term must go to zero at these two positions as well, which requires that the length L be exactly equal to a multiple of the half wavelength of the sine function:

$$\sqrt{\frac{P}{EI}} L = n\pi, \quad n = 1, 2, 3, \dots$$

The lowest value of P leading to the deformed shape corresponds to $n = 1$; the critical buckling load P_{cr} is then:

$$\boxed{P_{cr} = \frac{\pi^2 EI}{L^2}} \quad (11)$$

Note the dependency on L^2 , so the buckling load drops with the square of the length.

This strong dependency on length shows why crossbracing is so important in preventing buckling. If a brace is added at the beam's midpoint as shown in Fig. 7 to eliminate deflection

there, the buckling shape is forced to adopt a wavelength of L rather than $2L$. This is equivalent to making the beam half as long, which increases the critical buckling load by a factor of four.

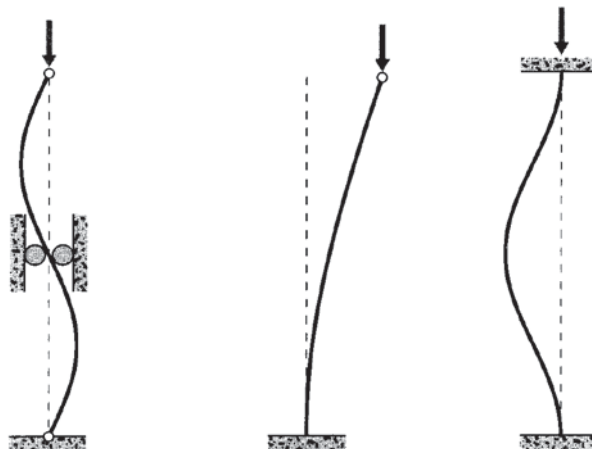


Figure 7: Effect of lateral support and end conditions on beam buckling.

Similar reasoning can be used to assess the result of having different support conditions. If for instance the beam is cantilevered at one end but unsupported at the other, its buckling shape will be a quarter sine wave. This is equivalent to making the beam twice as long as the case with both ends pinned, so the buckling load will go *down* by a factor of four. Cantilevering *both* ends forces a full-wave shape, with the same buckling load as the pinned beam with a midpoint support.

Shear stresses

Transverse loads bend beams by inducing axial tensile and compressive normal strains in the beam's x -direction, as discussed above. In addition, they cause shear effects that tend to slide vertical planes tangentially to one another as depicted in Fig. 8, much like sliding playing cards past one another. The stresses τ_{xy} associated with this shearing effect add up to the vertical shear force we have been calling V , and we now seek to understand how these stresses are distributed over the beam's cross section. The shear stress on vertical planes must be accompanied by an equal stress on horizontal planes since $\tau_{xy} = \tau_{yx}$, and these horizontal shearing stresses must become zero at the upper and lower surfaces of the beam unless a traction is applied there to balance them. Hence they must reach a maximum somewhere within the beam.

The variation of this horizontal shear stress with vertical position y can be determined by examining a free body of width dx cut from the beam a distance y above neutral axis as shown in Fig. 9. The moment on the left vertical face is $M(x)$, and on the right face it has increased to $M + dM$. Since the horizontal normal stresses are directly proportional to the moment ($\sigma_x = My/I$), any increment in moment dM over the distance dx produces an imbalance in the horizontal force arising from the normal stresses. This imbalance must be compensated by a shear stress τ_{xy} on the horizontal plane at y . The horizontal force balance is written as

$$\tau_{xy} b dx = \int_{A'} \frac{dM \xi}{I} dA'$$

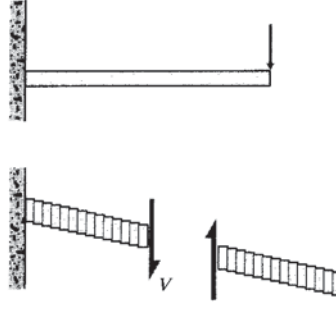


Figure 8: Shearing displacements in beam bending.

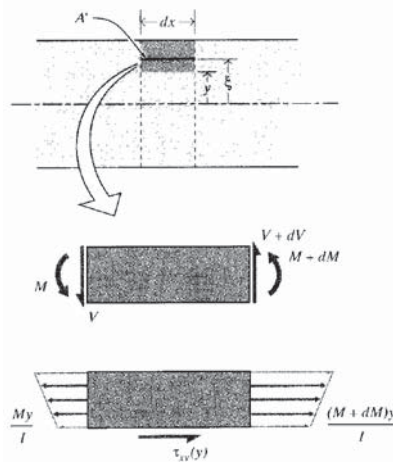


Figure 9: Shear and bending moment in a differential length of beam.

where b is the width of the beam at y , ξ is a dummy height variable ranging from y to the outer surface of the beam, and A' is the cross-sectional area between the plane at y and the outer surface. Using $dM = V dx$ from Eqn. 8 of Module 12, this becomes

$$\tau_{xy} = \frac{V}{Ib} \int_{A'} \xi dA' = \frac{VQ}{Ib} \quad (12)$$

where here $Q(y) = \int_{A'} \xi dA' = \bar{\xi}A'$ is the first moment of the area above y about the neutral axis.

The parameter $Q(y)$ is notorious for confusing persons new to beam theory. To determine it for a given height y relative to the neutral axis, begin by sketching the beam cross section, and draw a horizontal line at the position y at which Q is sought (Fig. 10 shows a rectangular beam of constant width b and height h for illustration). Note the area A' between this line and the outer surface (indicated by cross-hatching in Fig. 10). Now compute the distance $\bar{\xi}$ from the neutral axis to the centroid of A' . The parameter $Q(y)$ is the product of A' and $\bar{\xi}$; this is the first moment of the area A' with respect to the centroidal axis. For the rectangular beam, it is

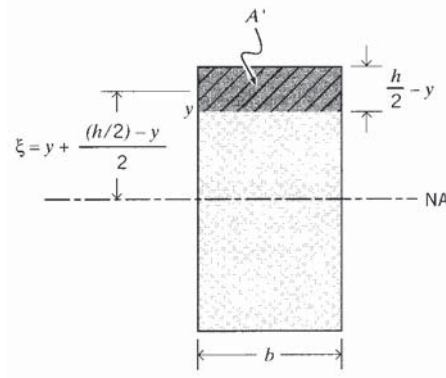


Figure 10: Section of a rectangular beam.

$$Q = A' \bar{\xi} = \left[b \left(\frac{h}{2} - y \right) \right] \left[y + \frac{1}{2} \left(\frac{h}{2} - y \right) \right] = \frac{b}{2} \left(\frac{h^2}{4} - y^2 \right)$$

Note that $Q(y)$, and therefore $\tau_{xy}(y)$ as well, is parabolic, being maximum at the neutral axis ($y = 0$) and zero at the outer surface ($y = h/2$). Using $I = bh^3/12$ for the rectangular beam, the maximum shear stress as given by Eqn. 12 is

$$\tau_{xy,max} = \tau_{xy}|_{y=0} = \frac{3V}{2bh}$$

(Keep in mind that the above two expressions for Q and $\tau_{xy,max}$ are for rectangular cross section only; sections of other shapes will have different results.) These shear stresses are most important in beams that are short relative to their height, since the bending moment usually increases with length and the shear force does not (see Prob. 11). One standard test for interlaminar shear strength² is to place a short beam in bending and observe the load at which cracks develop along the midplane.

Example 2

Since the normal stress is maximum where the horizontal shear stress is zero (at the outer fibers), and the shear stress is maximum where the normal stress is zero (at the neutral axis), it is often possible to consider them one at a time. However, the juncture of the web and the flange in I and T beams is often a location of special interest, since here both stresses can take on substantial values.

Consider the T beam seen previously in Example 1, and examine the location at point A shown in Fig. 11, in the web immediately below the flange. Here the width b in Eqn. 12 is the dimension labeled c ; since the beam is thin here the shear stress τ_{xy} will tend to be large, but it will drop dramatically in the flange as the width jumps to the larger value a . The normal stress at point A is computed from $\sigma_x = My/I$, using $y = d - \bar{y}$. This value will be almost as large as the outer-fiber stress if the flange thickness b is small compared with the web height d . The Mohr's circle for the stress state at point A would then have appreciable contributions from both σ_x and τ_{xy} , and can result in a principal stress larger than at either the outer fibers or the neutral axis.

This problem provides a good review of the governing relations for normal and shear stresses in beams, and is also a natural application for symbolic-manipulation computer methods. Using Maple software, we might begin by computing the location of the centroidal axis:

²“Apparent Horizontal Shear Strength of Reinforced Plastics by Short Beam Method,” ASTM D2344, American Society for Testing and Materials.

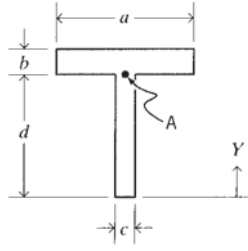


Figure 11: Section of T beam.

```
> ybar := ((d/2)*c*d) + ( (d+(b/2) )*a*b )/( c*d + a*b );
```

Here the “>” symbol is the Maple prompt, and the “;” is needed by Maple to end the command. The maximum shear force and bending moment (present at the wall) are defined in terms of the distributed load and the beam length as

```
> V := w*L;
> M := -(w*L)*(L/2);
```

For plotting purposes, it will be convenient to have a height variable Y measured from the bottom of the section. The relations for normal stress, shear stress, and the first principal stress are functions of Y ; these are defined using the Maple “procedure” command:

```
> sigx := proc (Y) -M*(Y-ybar)/Iz end;
> tauxy := proc (Y) V*Q(Y)/(Iz*B(Y) ) end;
> sigp1 := proc (Y) (sigx(Y)/2) + sqrt( (sigx(Y)/2)^2 + (tauxy(Y))^2 ) end;
```

The moment of inertia I_z is computed as

```
> I1 := (a*b^3)/12 + a*b* (d+(b/2)-ybar)^2;
> I2 := (c*d^3)/12 + c*d* ((d/2)-ybar)^2;
> Iz := I1+I2;
```

The beam width B is defined to take the appropriate value depending on whether the variable Y is in the web or the flange:

```
> B:= proc (Y) if Y<d then B:=c else B:=a fi end;
```

The command “fi” (“if” spelled backwards) is used to end an if-then loop. The function $Q(Y)$ is defined for the web and the flange separately:

```
> Q:= proc (Y) if Y<d then
>   int( (yy-ybar)*c,yy=Y..d) + int( (yy-ybar)*a,yy=d..(d+b) )
> else
>   int( (yy-ybar)*a,yy=Y.. (d+b) )
> fi end;
```

Here “int” is the Maple command for integration, and yy is used as the dummy height variable. The numerical values of the various parameters are defined as

```
> a:=3: b:=1/4: c:=1/4: d:=3-b: L:=8: w:=100:
```

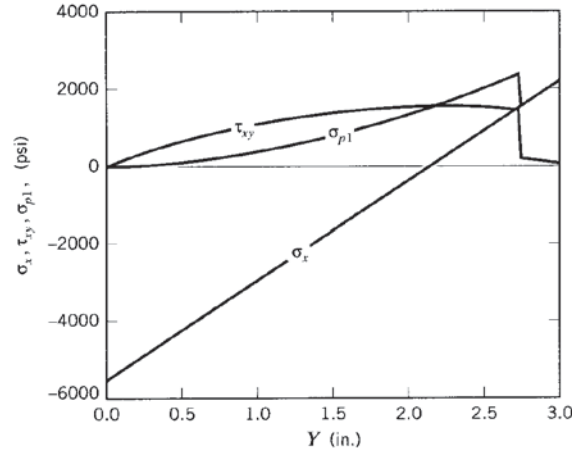


Figure 12: Stresses at the web-flange junction in a short cantilevered T beam subjected to uniform loading.

Finally, the stresses can be graphed using the Maple plot command

```
> plot({sigx,tauxy,sigp1},Y=0..3,sigx=-500..2500);
```

The resulting plot is shown in Fig. 12.

Example 3

In the previous example, we were interested in the variation of stress as a function of height in a beam of irregular cross section. Another common design or analysis problem is that of the variation of stress not only as a function of height but also of distance along the span dimension of the beam. The shear and bending moments $V(x)$ and $M(x)$ vary along this dimension, and so naturally do the stresses $\sigma_x(x, y)$ and $\tau_{xy}(x, y)$ that depend on them according to Eqns. 7 and 12.

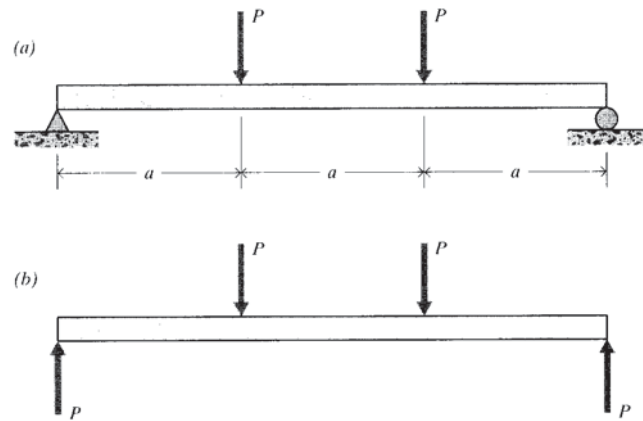


Figure 13: (a) Beam in four-point bending. (b) Free-body diagram.

Consider a short beam of rectangular cross section subjected to four-point loading as seen in Fig. 13. The loading, shear, and bending moment functions are:

$$q(x) = P\langle x \rangle_{-1} - P\langle x - a \rangle_{-1} - P\langle x - 2a \rangle_{-1} + P\langle x - 3a \rangle_{-1}$$

$$V(x) = - \int q(x) dx = -P\langle x \rangle^0 + P\langle x - a \rangle^0 + P\langle x - 2a \rangle^0 - P\langle x - 3a \rangle^0$$

$$M(x) = - \int V(x) dx = P\langle x \rangle^1 - P\langle x - a \rangle^1 - P\langle x - 2a \rangle^1 + P\langle x - 3a \rangle^1$$

The shear and normal stresses can be determined as functions of x and y directly from these functions, as well as such parameters as the principal stress. Since σ_y is zero everywhere, the principal stress is

$$\sigma_{p1} = \frac{\sigma_x}{2} + \sqrt{\left(\frac{\sigma_x}{2}\right)^2 + \tau_{xy}^2}$$

One way to visualize the x - y variation of σ_{p1} is by means of a 3D surface plot, which can be prepared easily by Maple. For the numerical values $P = 100, a = h = 10, b = 3$, we could use the expressions (Maple responses removed for brevity):

```
> # use Heaviside for singularity functions
> readlib(Heaviside);
> sf := proc(x,a,n) (x-a)^n * Heaviside(x-a) end;
> # define shear and bending moment functions
> V := (x) -> -P*sfn(x,0,0)+P*sfn(x,a,0)+P*sfn(x,2*a,0)-P*sfn(x,3*a,0);
> M := (x) -> P*sfn(x,0,1)-P*sfn(x,a,1)-P*sfn(x,2*a,1)+P*sfn(x,3*a,1);
> # define shear stress function
> tau := V(x)*Q/(Iz*b);
> Q := (b/2)*( h^2/4 - y^2);
> Iz := b*h^3/12;
> # define normal stress function
> sig := M(x)*y/Iz;
> # define principal stress
> sigp := (sig/2) + sqrt( (sig/2)^2 + tau^2 );
> # define numerical parameters
> P:=100;a:=10;h:=10;b:=3;
> # make plot
> plot3d(sigp,x=0..3*a,y=-h/2 .. h/2);
```

The resulting plot is shown in Fig. 14. The dominance of the parabolic shear stress is evident near the beam ends, since here the shear force is at its maximum value but the bending moment is small (plot the shear and bending moment diagrams to confirm this). In the central part of the beam, where $a < x < 2a$, the shear force vanishes and the principal stress is governed only by the normal stress σ_x , which varies linearly from the beam's neutral axis. The first principal stress is zero in the compressive lower part of this section, since here the normal stress σ_x is negative and the right edge of the Mohr's circle must pass through the zero value of the other normal stress σ_y . Working through the plot of Fig. 14 is a good review of the beam stress formulas.

Problems

1. Derive the composite area theorem for determining the centroid of a compound area.

$$\bar{y} = \frac{\sum_i A_i \bar{y}_i}{\sum_i A_i}$$

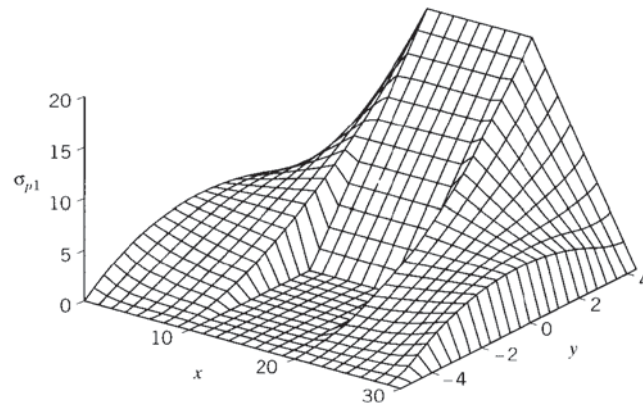
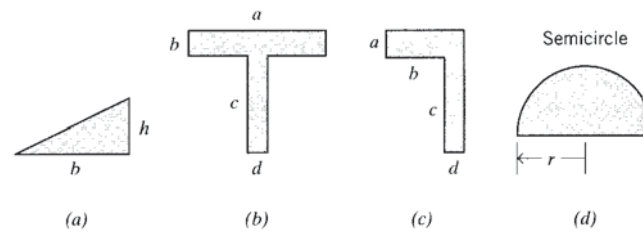


Figure 14: Variation of principal stress σ_{p1} in four-point bending.

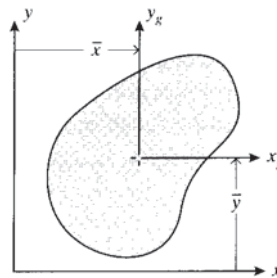


Prob. 2

2. (a)–(d) Locate the centroids of the areas shown.
3. Derive the “parallel-axis theorem” for moments of inertia of a plane area:

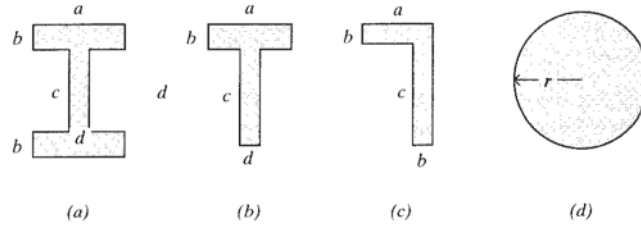
$$I_x = I_{xg} + A\bar{y}^2$$

$$I_y = I_{yg} + A\bar{x}^2$$



Prob. 3

4. (a)–(d) Determine the moment of inertia relative to the horizontal centroidal axis of the areas shown.



Prob. 4

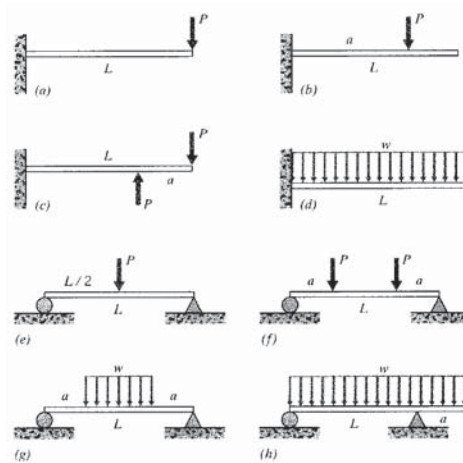
5. Show that the moment of inertia transforms with respect to axis rotations exactly as does the stress:

$$I_{x'} = I_x \cos^2 \theta + I_y \sin^2 \theta - 2I_{xy} \sin \theta \cos \theta$$

where I_x and I_y are the moments of inertia relative to the x and y axes respectively and I_{xy} is the *product of inertia* defined as

$$I_{xy} = \int_A xy dA$$

6. (a)–(h) Determine the maximum normal stress σ_x in the beams shown here, using the values (as needed) $L = 25$ in, $a = 5$ in, $w = 10$ lb/in, $P = 150$ lb. Assume a rectangular cross-section of width $b = 1$ in and height $h = 2$ in.



Prob. 6

7. Justify the statement in ASTM test D790, “Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials,” which reads:

When a beam of homogeneous, elastic material is tested in flexure as a simple beam supported at two points and loaded at the midpoint, the maximum stress in the outer fibers occurs at midspan. This stress may be calculated for any point on the load-deflection curve by the following equation:

$$S = 3PL/2bd^2$$

where S = stress in the outer fibers at midspan, MPa; P = load at a given point on the load-deflection curve; L = support span, mm; b = width of beam tested, mm; and d = depth of beam tested, mm.

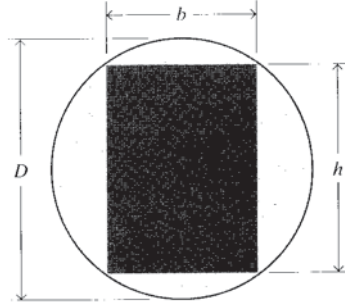
8. Justify the statement in ASTM test D790, "Standard Test Methods for Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials," which reads:

The tangent modulus of elasticity, often called the "modulus of elasticity," is the ratio, within the elastic limit of stress to corresponding strain and shall be expressed in megapascals. It is calculated by drawing a tangent to the steepest initial straight-line portion of the load-deflection curve and using [the expression:]

$$E_b = L^3 m / 4bd^3$$

where E_b = modulus of elasticity in bending, MPa; L = support span, mm; d = depth of beam tested, mm; and m = slope of the tangent to the initial straight-line portion of the load-deflection curve, N/mm of deflection.

9. A rectangular beam is to be milled from circular stock as shown. What should be the ratio of height to width (b/h) to as to minimize the stresses when the beam is put into bending?

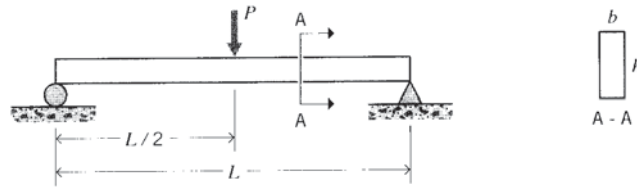


Prob. 9

10. (a)–(h) Determine the maximum shear τ_{xy} in the beams of Prob. 6, , using the values (as needed) $L = 25$ in, $a = 5$ in, $w = 10$ lb/in, $P = 150$ lb. Assume a rectangular cross-section of width $b = 1$ in and height $h = 2$ in.
11. Show that the ratio of maximum shearing stress to maximum normal stress in a beam subjected to 3-point bending is

$$\frac{\tau}{\sigma} = \frac{h}{2L}$$

Hence the importance of shear stress increases as the beam becomes shorter in comparison with its height.



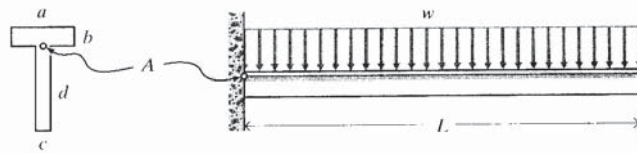
Prob. 11

12. Read the ASTM test D4475, “Standard Test Method for Apparent Horizontal Shear Strength of Pultruded Reinforced Plastic Rods By The Short-Beam Method,” and justify the expression given there for the apparent shear strength:

$$S = 0.849P/d^2$$

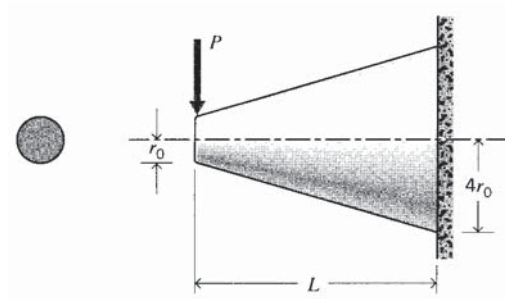
where S = apparent shear strength, N/m², (or psi); P = breaking load, N, (or lbf); and d = diameter of specimen, m (or in.).

13. For the T beam shown here, with dimensions $L = 3$, $a = 0.05$, $b = 0.005$, $c = 0.005$, $d = 0.7$ (all in m) and a loading distribution of $w = 5000$ N/m, determine the principal and maximum shearing stress at point A.



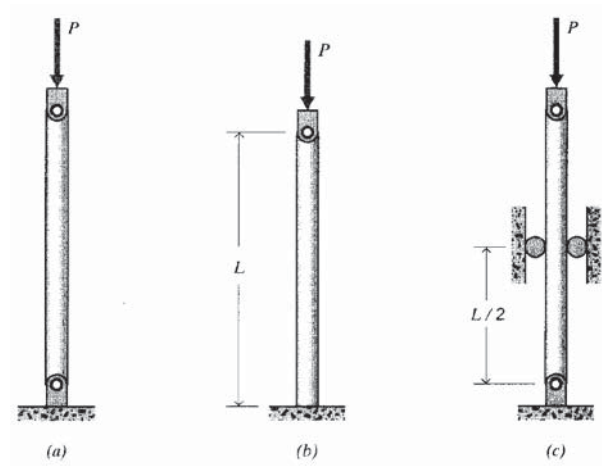
Prob. 13

14. Determine the maximum normal stress in a cantilevered beam of circular cross section whose radius varies linearly from $4r_0$ to r_0 in a distance L , loaded with a force P at the free end.



Prob. 14

15. A carbon steel column has a length $L = 1$ m and a circular cross section of diameter $d = 20$ mm. Determine the critical buckling load P_c for the case of (a) both ends pinned, (b) one end cantilevered, (c) both ends pinned but supported laterally at the midpoint.



Prob. 15

16. A carbon steel column has a length $L = 1$ m and a circular cross section. Determine the diameter d at which the column has an equal probability of buckling or yielding in compression.

Beam Displacements

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Introduction

We want to be able to predict the deflection of beams in bending, because many applications have limitations on the amount of deflection that can be tolerated. Another common need for deflection analysis arises from materials testing, in which the transverse deflection induced by a bending load is measured. If we know the relation expected between the load and the deflection, we can “back out” the material properties (specifically the modulus) from the measurement. We will show, for instance, that the deflection at the midpoint of a beam subjected to “three-point bending” (beam loaded at its center and simply supported at its edges) is

$$\delta_P = \frac{PL^3}{48EI}$$

where the length L and the moment of inertia I are geometrical parameters. If the ratio of δ_P to P is measured experimentally, the modulus E can be determined. A stiffness measured this way is called the *flexural modulus*.

There are a number of approaches to the beam deflection problem, and many texts spend a good deal of print on this subject. The following treatment outlines only a few of the more straightforward methods, more with a goal of understanding the general concepts than with developing a lot of facility for doing them manually. In practice, design engineers will usually consult handbook tabulations of deflection formulas as needed, so even before the computer age many of these methods were a bit academic.

Multiple integration

In Module 12, we saw how two integrations of the loading function $q(x)$ produces first the shear function $V(x)$ and then the moment function $M(x)$:

$$V = - \int q(x) dx + c_1 \tag{1}$$

$$M = - \int V(x) dx + c_2 \tag{2}$$

where the constants of integration c_1 and c_2 are evaluated from suitable boundary conditions on V and M . (If singularity functions are used, the boundary conditions are included explicitly and the integration constants c_1 and c_2 are identically zero.) From Eqn. 6 in Module 13, the curvature

$v_{,xx}(x)$ is just the moment divided by the section modulus EI . Another two integrations then give

$$v_{,x}(x) = \frac{1}{EI} \int M(x) dx + c_3 \quad (3)$$

$$v(x) = \int v_{,x}(x) dx + c_4 \quad (4)$$

where c_3 and c_4 are determined from boundary conditions on slope or deflection.

Example 1

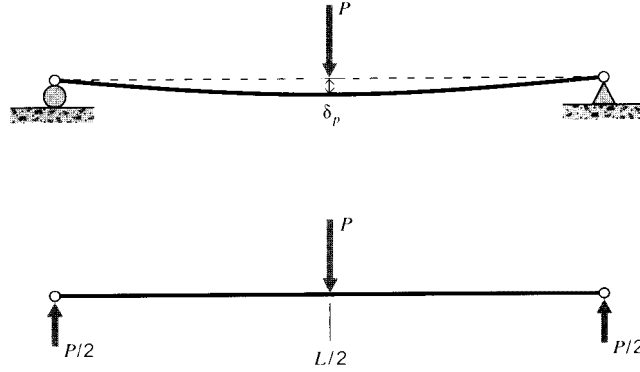


Figure 1: Three-point bending.

As an illustration of this process, consider the case of “three-point bending” shown in Fig. 1. This geometry is often used in materials testing, as it avoids the need to clamp the specimen to the testing apparatus. If the load P is applied at the midpoint, the reaction forces at A and B are equal to half the applied load. The loading function is then

$$q(x) = \frac{P}{2} \langle x \rangle_{-1} - P \langle x - \frac{L}{2} \rangle_{-1}$$

Integrating according to the above scheme:

$$V(x) = -\frac{P}{2} \langle x \rangle^0 + P \langle x - \frac{L}{2} \rangle^0$$

$$M(x) = \frac{P}{2} \langle x \rangle^1 - P \langle x - \frac{L}{2} \rangle^1 \quad (5)$$

$$EI v_{,x}(x) = \frac{P}{4} \langle x \rangle^2 - \frac{P}{2} \langle x - \frac{L}{2} \rangle^2 + c_3$$

From symmetry, the beam has zero slope at the midpoint. Hence $v_{,x} = 0$ @ $x = L/2$, so c_3 can be found to be $-PL^2/16$. Integrating again:

$$EI v(x) = \frac{P}{12} \langle x \rangle^3 - \frac{P}{6} \langle x - \frac{L}{2} \rangle^3 - \frac{PL^2 x}{16} + c_4$$

The deflection is zero at the left end, so $c_4 = 0$. Rearranging, the beam deflection is given by

$$v = \frac{P}{48EI} \left[4x^3 - 3L^2x - 8\left(x - \frac{L}{2}\right)^3 \right] \quad (6)$$

The maximum deflection occurs at $x = L/2$, which we can evaluate just before the singularity term activates. Then

$$\delta_{\max} = \frac{PL^3}{48EI} \quad (7)$$

This expression is much used in flexural testing, and is the example used to begin this module.

Before the loading function $q(x)$ can be written, the reaction forces at the beam supports must be determined. If the beam is statically determinate, as in the above example, this can be done by invoking the equations of static equilibrium. Static determinacy means only two reaction forces or moments can be present, since we have only a force balance in the direction transverse to the beam axis and one moment equation available. A simply supported beam (one resting on only two supports) or a simply cantilevered beam are examples of such determinate beams; in the former case there is one reaction force at each support, and in the latter case there is one transverse force and one moment at the clamped end.

Of course, there is no stringent engineering reason to limit the number of beam supports to those sufficient for static equilibrium. Adding “extra” supports will limit deformations and stresses, and this will often be worthwhile in spite of the extra construction expense. But the analysis is now a bit more complicated, since not all of the unknown reactions can be found from the equations of static equilibrium. In these statically indeterminate cases it will be necessary to invoke geometrical constraints to develop enough equations to solve the problem.

This is done by writing the slope and deflection equations, carrying the unknown reaction forces and moments as undetermined parameters. The slopes and deflections are then set to their known values at the supports, and the resulting equations solved for the unknowns. If for instance a beam is resting on three supports, there will be three unknown reaction forces, and we will need a total of five equations: three for the unknown forces and two more for the constants of integration that arise when the slope and deflection equations are written. Two of these equations are given by static equilibrium, and three more are obtained by setting the deflections at the supports to zero. The following example illustrates the procedure, which is straightforward although tedious if done manually.

Example 2

Consider a triply-supported beam of length $L = 15$ as shown in Fig. 2, carrying a constant uniform load of $w = -10$. There are not sufficient equilibrium equations to determine the reaction forces R_a , R_b , and R_c , so these are left as unknowns while multiple integration is used to develop a deflection equation:

$$\begin{aligned} q(x) &= R_a \langle x \rangle_{-1} + R_b \langle x - 7.5 \rangle_{-1} + R_c \langle x - 15 \rangle_{-1} - 10 \langle x \rangle^0 \\ V(x) &= - \int q(x) dx = -R_a \langle x \rangle^0 - R_b \langle x - 7.5 \rangle^0 - R_c \langle x - 15 \rangle^0 + 10 \langle x \rangle^1 \\ M(x) &= - \int V(x) dx = R_a \langle x \rangle^1 + R_b \langle x - 7.5 \rangle^1 + R_c \langle x - 15 \rangle^1 - \frac{10}{2} \langle x \rangle^2 \\ EI y'(x) &= \int M(x) dx = \frac{R_a}{2} \langle x \rangle^2 + \frac{R_b}{2} \langle x - 7.5 \rangle^2 + \frac{R_c}{2} \langle x - 15 \rangle^2 - \frac{10}{6} \langle x \rangle^3 + c_1 \end{aligned}$$

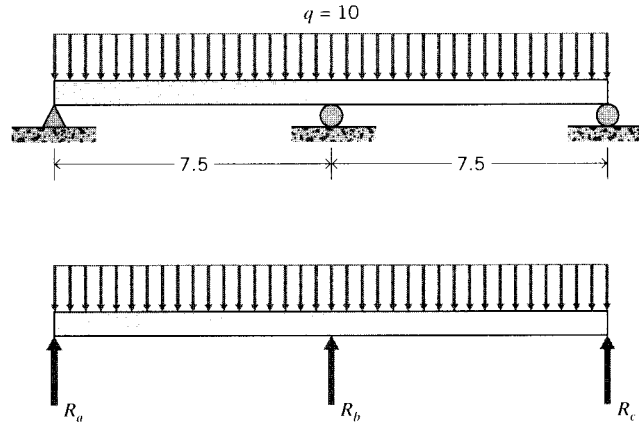


Figure 2: Uniformly loaded beam resting on three supports.

$$EIy(x) = \int EIy'(x) dx = \frac{R_a}{6}\langle x \rangle^3 + \frac{R_b}{6}\langle x - 7.5 \rangle^3 + \frac{R_c}{6}\langle x - 15 \rangle^3 - \frac{10}{24}\langle x \rangle^4 + c_1x + c_2$$

These equations have 5 unknowns: R_a , R_b , R_c , c_1 , and c_2 . These must be obtained from the two equilibrium equations

$$\sum F_y = 0 = R_a + R_b + R_c - qL$$

$$\sum M_a = 0 = qL\frac{L}{2} - R_b\frac{L}{2} - R_cL$$

and the three known zero displacements at the supports

$$y(0) = y(L/2) = y(L) = 0$$

Although the process is straightforward, there is a lot of algebra to wade through. Statically indeterminate beams tend to generate tedious mathematics, but fortunately this can be reduced greatly by modern software. Follow how easily this example is handled by the Maple V package (some of the Maple responses removed for brevity):

```
> # read the library containing the Heaviside function
> readlib(Heaviside);
> # use the Heaviside function to define singularity functions;
> #   sfm(x,a,n) is same as <x-a>^n
> sfm := proc(x,a,n) (x-a)^n * Heaviside(x-a) end;
> # define the deflection function:
> y := (x)-> (Ra/6)*sfm(x,0,3)+(Rb/6)*sfm(x,7.5,3)+(Rc/6)*sfm(x,15,3)
>   -(10/24)*sfm(x,0,4)+c1*x+c2;
> # Now define the five constraint equations; first vertical equilibrium:
> eq1 := 0=Ra+Rb+Rc-(10*15);
> # rotational equilibrium:
> eq2 := 0=(10*15*7.5)-Rb*7.5-Rc*15;
> # Now the three zero displacements at the supports:
> eq3 := y(0)=0;
> eq4 := y(7.5)=0;
> eq5 := y(15)=0;
> # set precision; 4 digits is enough:
```

```

> Digits:=4;
> # solve the 5 equations for the 5 unknowns:
> solve({eq1,eq2,eq3,eq4,eq5},{Ra,Rb,Rc,c1,c2});
      {c2 = 0, c1 = -87.82, Rb = 93.78, Ra = 28.11, Rc = 28.11}
> # assign the known values for plotting purposes:
> c1:=-87.82;c2:=0;Ra:=28.11;Rb:=93.78;Rc:=28.11;
> # the equation of the deflection curve is:
> y(x);
      3              3
      4.686 x  Heaviside(x) + 15.63 (x - 7.5)  Heaviside(x - 7.5)
      3              4
      + 4.686 (x - 15)  Heaviside(x - 15) - 5/12 x  Heaviside(x) - 87.82 x
> # plot the deflection curve:
> plot(y(x),x=0..15);
> # The maximum deflection occurs at the quarter points:
> y(15/4);
      -164.7

```

The plot of the deflection curve is shown in Fig. 3.

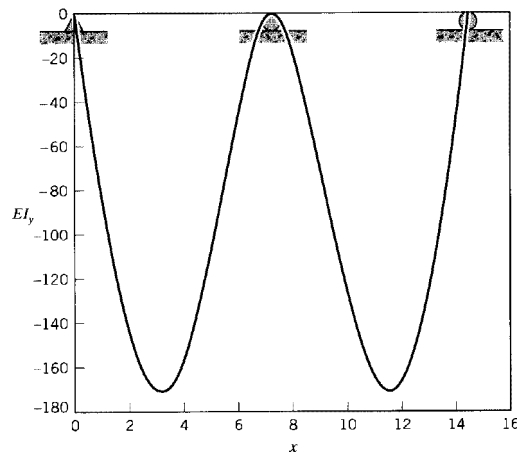


Figure 3: Deflection curve $EIy(x)$ for uniformly loaded triply-supported beam (Note difference in horizontal and vertical scales).

Energy method

The strain energy in bending as given by Eqn. 8 of Module 13 can be used to find deflections, and this may be more convenient than successive integration if the deflection at only a single point is desired. Castigliano's Theorem gives the deflection congruent to a load P as

$$\delta_P = \frac{\partial U}{\partial P} = \frac{\partial}{\partial P} \int_L \frac{M^2 dx}{2EI}$$

It is usually more convenient to do the differentiation before the integration, since this lowers the order of the expression in the integrand:

$$\delta_P = \int_L \frac{M}{EI} \frac{\partial M}{\partial P} dx$$

where here E and I are assumed not to vary with x .

The shear contribution to bending can be obtained similarly. Knowing the shear stress $\tau = VQ/Ib$ (omitting the xy subscript on τ for now), the strain energy due to shear U_s can be written

$$U_s = \int_V \frac{\tau^2}{2G} dV = \int_L \frac{V^2}{2GI} \left[\int_A \frac{Q^2}{E^2} dA \right] dx$$

The integral over the cross-sectional area A is a purely geometrical factor, and we can write

$$U_s = \int_L \frac{V^2 f_s}{2GA} dA \quad (8)$$

where the f_s is a dimensionless *form factor for shear* defined as

$$f_s = \frac{A}{I^2} \int_A \frac{Q^2}{b^2} dA \quad (9)$$

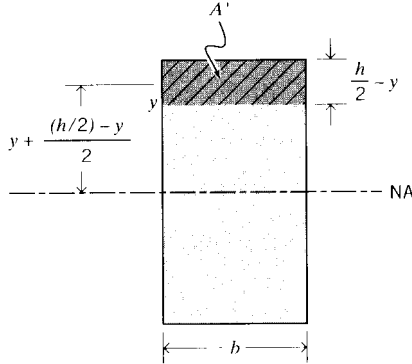


Figure 4: Rectangular beam section.

Evaluating f_s for rectangular sections for illustration (see Fig. 4), we have in that case

$$\begin{aligned} A &= bh, \quad I = \frac{bh^3}{12} \\ Q &= \left[y + \frac{(h/2) - y}{2} \right] \left[b \left(\frac{h}{2} - y \right) \right] \\ f_s &= \frac{(bh)}{(bh^3/12)^2} \int_{-h/2}^{h/2} \frac{1}{b^2} Q^2 dy = \frac{6}{5} \end{aligned}$$

Hence f_s is the same for all rectangular sections, regardless of their particular dimensions. Similarly, it can be shown (see Prob. 3) that for solid circular sections $f_s = 10/9$ and for hollow circular sections $f_s = 2$.

Example 3

If for instance we are seeking the deflection under the load P in the three-point bending example done earlier, we can differentiate the moment given in Eqn. 5 to obtain

$$\frac{\partial M}{\partial P} = \frac{1}{2}\langle x \rangle^1 - \langle x - \frac{L}{2} \rangle^1$$

Then

$$\delta_P = \frac{1}{EI} \int_L \left(\frac{P}{2}\langle x \rangle^1 - P\langle x - \frac{L}{2} \rangle^1 \right) \left(\frac{1}{2}\langle x \rangle^1 - \langle x - \frac{L}{2} \rangle^1 \right) dx$$

Expanding this and adjusting the limits of integration to account for singularity functions that have not been activated:

$$\begin{aligned} \delta_P &= \frac{P}{EI} \left\{ \int_0^L \frac{x^2}{4} dx + \int_{L/2}^L \left[-x \left(x - \frac{L}{2} \right) + \left(x - \frac{L}{2} \right)^2 \right] dx \right\} \\ &= -\frac{PL^3}{48EI} \end{aligned}$$

as before.

The contribution of shear to the deflection can be found by using $V = P/2$ in the equation for strain energy. For the case of a rectangular beam with $f_s = 6/5$ we have:

$$\begin{aligned} U_s &= \frac{(P/2)^2(6/5)}{2GA} L \\ \delta_{P,s} &= \frac{\partial U_s}{\partial P} = \frac{6PL}{20GA} \end{aligned}$$

The shear contribution can be compared with the bending contribution by replacing A with $12I/h^2$ (since $A = bh$ and $I = bh^3/12$). Then the ratio of the shear to bending contributions is

$$\frac{PLh^2/40GI}{PL^3/24EI} = \frac{3h^2E}{5L^2G}$$

Hence the importance of the shear term scales as $(h/L)^2$, i.e. quadratically as the span-to-depth ratio.

The energy method is often convenient for systems having complicated geometries and combined loading. For slender shafts transmitting axial, torsional, bending and shearing loads the strain energy is

$$\boxed{U = \int_L \left(\frac{P^2}{2EA} + \frac{T^2}{2GJ} + \frac{M^2}{2EI} + \frac{V^2 f_s}{2GA} \right) dx} \quad (10)$$

Example 4

Consider a cantilevered circular beam as shown in Fig. 5 that tapers from radius r_1 to r_2 over the length L . We wish to determine the deflection caused by a force F applied to the free end of the beam, at an angle θ from the horizontal. Turning to Maple to avoid the algebraic tedium, the dimensional parameters needed in Eqn. 10 are defined as:

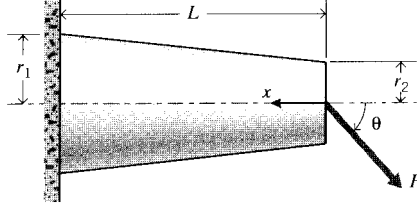


Figure 5: Tapered circular beam.

```
> r := proc (x) r1 + (r2-r1)*(x/L) end;
> A := proc (r) Pi*(r(x))^2 end;
> Iz := proc (r) Pi*(r(x))^4 /4 end;
> Jp := proc (r) Pi*(r(x))^4 /2 end;
```

where $r(x)$ is the radius, $A(r)$ is the section area, I_z is the rectangular moment of inertia, and J_p is the polar moment of inertia. The axial, bending, and shear loads are given in terms of F as

```
> P := F* cos(theta);
> V := F* sin(theta);
> M := proc (x) -F* sin(theta) * x end;
```

The strain energies corresponding to tension, bending and shear are

```
> U1 := P^2/(2*E*A(r));
> U2 := (M(x))^2/(2*E*Iz(r));
> U3 := V^2*(10/9)/(2*G*A(r));
> U := int( U1+U2+U3, x=0..L);
```

Finally, the deflection congruent to the load F is obtained by differentiating the total strain energy:

```
> dF := diff(U,F);
```

The result of these manipulations yields

$$\delta_F = \frac{LF [12 L^2 G - 12 GL^2 \cos^2 \theta + 9 Gr_2^2 \cos^2 \theta + 10 r_2^2 E - 10 r_2^2 E \cos^2 \theta]}{9 r_1 r_2^3 E \pi G}$$

This displacement is in the direction of the applied force F ; the horizontal and vertical deflections of the end of the beam are then

$$\delta_x = \delta_F \cos \theta$$

$$\delta_y = \delta_F \sin \theta$$

Superposition

In practice, many beams will be loaded in a complicated manner consisting of several concentrated or distributed loads acting at various locations along the beam. Although these multiple-load cases can be solved from scratch using the methods described above, it is often easier to solve the problem by superposing solutions of simpler problems whose solutions are tabulated. Fig. 6 gives an abbreviated collection of deflection formulas¹ that will suffice for many problems. The superposition approach is valid since the governing equations are linear; hence the response to a combination of loads is the sum of the responses that would be generated by each separate load acting alone.

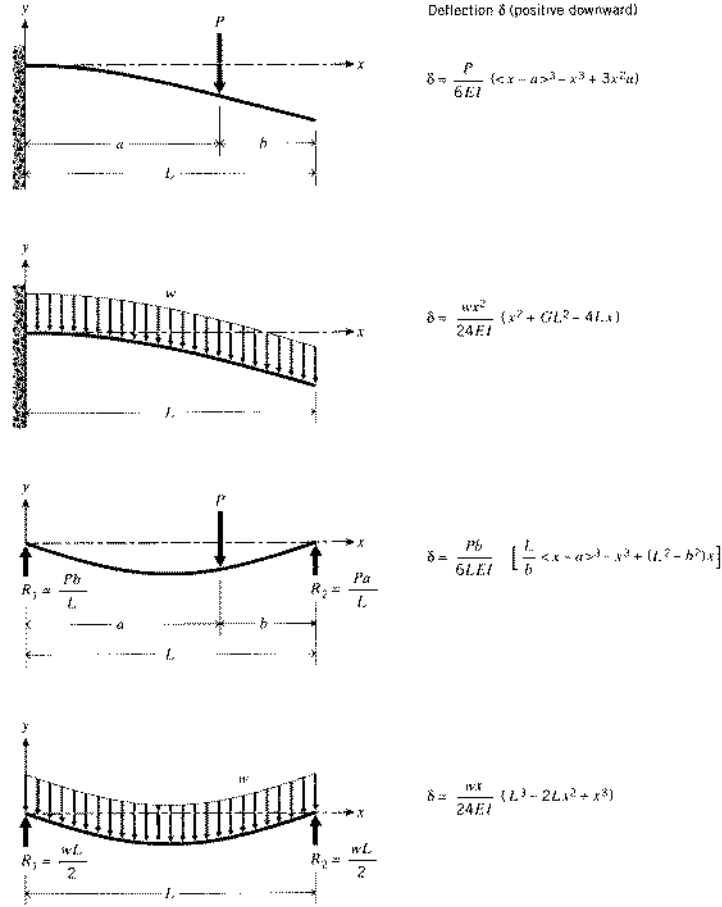


Figure 6: Deflections for cantilevered and simply-supported beams, under concentrated and distributed loading.

Example 5

We wish to find the equation of the deflection curve for a simply-supported beam loaded in symmetric four-point bending as shown in Fig. 7. From Fig. 6, the deflection of a beam with a single load at a

¹A more exhaustive listing is available in W.C. Young, *Roark's Formulas for Stress and Strain*, McGraw-Hill, New York, 1989.

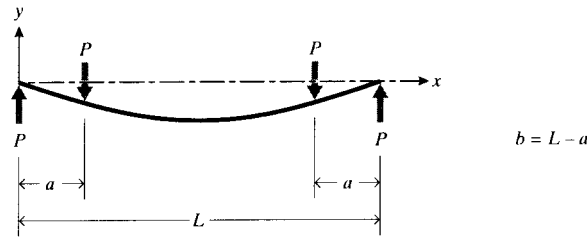


Figure 7: Four-point bending.

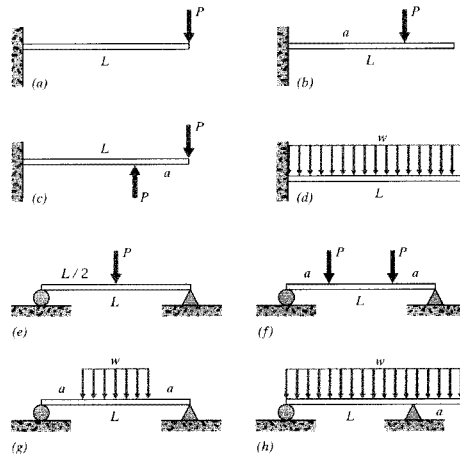
distance a from the left end is $\delta(x) = \frac{Pb}{6LEI} \left[\frac{L}{b} \langle x - a \rangle^3 - x^3 + (L^2 - b^2) x \right]$. Our present problem is just two such loads acting simultaneously, so we have

$$\delta(x) = \frac{P(L-a)}{6LEI} \left[\frac{L}{L-a} \langle x - a \rangle^3 - x^3 + (L^2 - (L-a)^2) x \right] \\ + \frac{Pa}{6LEI} \left[\frac{L}{a} \langle x - (L-a) \rangle^3 - x^3 + (L^2 - a^2) x \right]$$

In some cases the designer may not need the entire deflection curve, and superposition of tabulated results for maximum deflection and slope is equally valid.

Problems

- (a)–(h) Write expressions for the slope and deflection curves of the beams shown here.

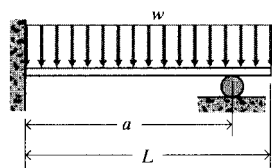


Prob. 1

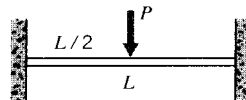
- (a)–(h) Use MapleV (or other) software to plot the slope and deflection curves for the beams in Prob. 1, using the values (as needed) $L = 25$ in, $a = 15$ in, $w = 10$ lb/in, $P = 150$ lb.
- Show that the shape factor for shear for a circular cross section is

$$f_s = \frac{A}{I^2} \int_A \frac{Q}{b^2} dA = \frac{10}{9}$$

4. (a)–(b) Determine the deflection curves for the beams shown here. Plot these curves for the values (as needed) $L = 25$ in, $a = 5$ in, $w = 10$ lb/in, $P = 150$ lb.



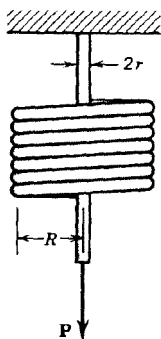
(a)



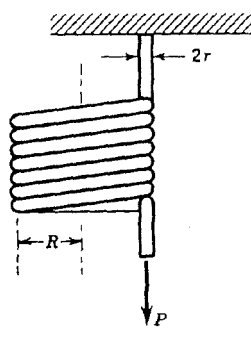
(b)

Prob. 4

5. (a) Determine the deflection of a coil spring under the influence of an axial force F , including the contribution of bending, direct shear, and torsional shear effects. Using $r = 1$ mm and $R = 10$ mm, compute the relative magnitudes of the three contributions.
- (b) Repeat the solution in (a), but take the axial load to be placed at the outer radius of the coil.



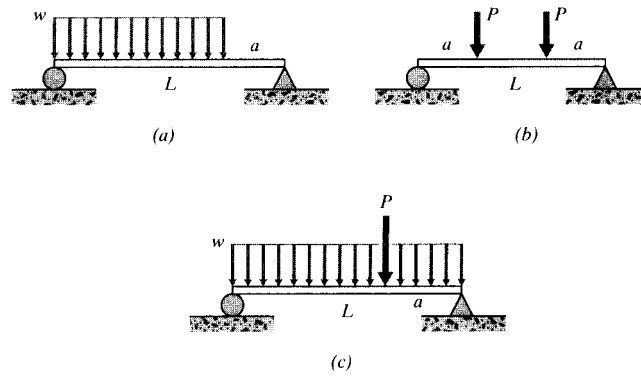
(a)



(b)

Prob. 5

6. (a)–(c) Use the method of superposition to write expressions for the deflection curve $\delta(x)$ for the cases shown here.



Prob. 6